



INNOVATION. AUTOMATION. ANALYTICS

Global Disaster Response Analysis

→ About Me

Background: B.Tech Graduate

Goal: Become a Data Analyst / Data Scientist

Interest: Data Analysis, Visualization, Business Insights

Experience: Fresher

- **Why do you want to learn Data Analytics?**

I want to learn Data Analytics to analyze data, identify patterns, and convert raw data into meaningful insights for better business decisions.

- **LinkedIn & GitHub**

LinkedIn : <https://www.linkedin.com/in/sindhu-sri-chattu-a9005a2b4/>

GitHub : <https://github.com/SindhusriChattu>

Dataset Overview

- The dataset contains 50,000 rows and 12 columns representing global disaster events.
- Columns include: Date, Country, Disaster Type, Severity Index, Casualties, Economic Loss (USD), Response Time (Hours), Aid Amount (USD), Response Efficiency Score, Recovery Days, Latitude, and Longitude.
- The dataset consists of 3 categorical and 9 numerical features used for exploratory data analysis.
- The data is analyzed to identify patterns, distributions, and relationships among variables.

Simple Description of each Column

- **Date:** The date on which the disaster event was recorded.
- **Country:** The country where the disaster occurred.
- **Disaster Type:** The category of disaster, such as earthquake, flood, or cyclone.
- **Severity Index:** A numerical score indicating the intensity or severity of the disaster.
- **Casualties:** The number of people injured or killed due to the disaster.
- **Economic Loss (USD):** The estimated financial loss caused by the disaster in US dollars.
- **Response Time (Hours):** The time taken by emergency services to respond after the disaster occurred.
- **Aid Amount (USD):** The total financial aid provided for disaster relief in US dollars.
- **Response Efficiency Score:** A score representing how effective the disaster response efforts were.
- **Recovery Days:** The number of days required for recovery and restoration after the disaster.
- **Latitude:** The geographical latitude of the disaster location.
- **Longitude:** The geographical longitude of the disaster location.

Reading Dataset

```
df=pd.read_csv(r"C:\Users\ch.sindhu sri\OneDrive\Pictures\Documents\OneDrive\Desktop\Inno Pandas\global_disaster_response_2018_2024 (1).csv")
df
```

	date	country	disaster_type	severity_index	casualties	economic_loss_usd	response_time_hours	aid_amount_usd	response_efficiency_score	recovery_days	latitude	longitude
0	2021-01-31	Brazil	Earthquake	5.99	111	7934365.71	15.62	271603.79	83.21	67	-30.613	-122.557
1	2018-12-23	Brazil	Extreme Heat	6.53	100	8307648.99	5.03	265873.81	96.18	55	10.859	-159.194
2	2020-08-10	India	Hurricane	1.55	22	765136.99	32.54	49356.49	60.40	22	0.643	-160.978
3	2022-09-15	Indonesia	Extreme Heat	4.55	94	1308251.31	7.83	237512.88	86.41	47	-33.547	30.350
4	2022-09-28	United States	Wildfire	3.80	64	2655864.36	21.90	188910.69	72.81	42	-19.170	-117.137
...	...	...	...	...	...	...	...	...	...	...	...	...
49995	2019-05-14	Chile	Landslide	5.50	78	3711240.93	8.45	305020.35	94.27	55	12.976	-25.680
49996	2020-10-30	United States	Wildfire	7.76	165	12072842.65	1.00	363881.25	95.46	76	57.265	-147.346
49997	2019-04-27	Turkey	Flood	4.90	130	1805859.70	5.14	280665.61	86.67	47	15.217	-27.856
49998	2022-10-09	Greece	Storm Surge	3.35	82	3176085.56	19.22	80331.23	84.75	32	-44.002	1.923
49999	2023-01-12	South Africa	Drought	5.03	129	2933495.70	9.08	354096.99	97.16	50	13.084	-81.048

Basic Checks

Columns Present in Dataset

```
df.columns

Index(['date', 'country', 'disaster_type', 'severity_index', 'casualties',
      'economic_loss_usd', 'response_time_hours', 'aid_amount_usd',
      'response_efficiency_score', 'recovery_days', 'latitude', 'longitude'],
      dtype='object')
```

Structure of the Dataset

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   date                  50000 non-null object
1   country               50000 non-null object
2   disaster_type         50000 non-null object
3   severity_index        50000 non-null float64
4   casualties            50000 non-null int64
5   economic_loss_usd     50000 non-null float64
6   response_time_hours   50000 non-null float64
7   aid_amount_usd        50000 non-null float64
8   response_efficiency_score 50000 non-null float64
9   recovery_days         50000 non-null int64
10  latitude              50000 non-null float64
11  longitude             50000 non-null float64
dtypes: float64(7), int64(2), object(3)
memory usage: 4.6+ MB
```

Checking Summary Statistics of Numerical Column

df.describe()									
	severity_index	casualties	economic_loss_usd	response_time_hours	aid_amount_usd	response_efficiency_score	recovery_days	latitude	longitude
count	50000.000000	50000.000000	5.000000e+04	50000.000000	5.000000e+04	50000.000000	50000.000000	50000.000000	50000.000000
mean	5.015769	100.591140	5.068593e+06	12.183027	2.500003e+05	87.574025	49.682560	0.231205	0.089889
std	1.942843	65.052064	3.268541e+06	9.259081	1.432275e+05	10.188961	20.098944	34.759662	98.380208
min	1.000000	0.000000	5.273900e+02	1.000000	1.660000e+01	29.750000	2.000000	-59.994000	-169.998000
25%	3.660000	51.000000	2.585513e+06	6.270000	1.429663e+05	83.060000	36.000000	-29.848250	-85.551750
50%	4.990000	91.000000	4.548351e+06	10.510000	2.305365e+05	89.180000	49.000000	0.267500	0.390500
75%	6.340000	138.000000	6.950615e+06	15.450000	3.352259e+05	94.700000	63.000000	30.488250	85.665000
max	10.000000	524.000000	2.445624e+07	63.100000	1.126465e+06	100.000000	112.000000	59.997000	169.999000

Checking Summary Statistics of Categorical Column

df.describe(include='O')			
	date	country	disaster_type
count	50000	50000	50000
unique	2557	20	10
top	2023-08-29	Brazil	Landslide
freq	36	2591	5130

Data Quality Checks

Checking Missing Values

```
df.isna().sum()
```

```
date                0
country             0
disaster_type       0
severity_index      0
casualties          0
economic_loss_usd   0
response_time_hours 0
aid_amount_usd      0
response_efficiency_score 0
recovery_days       0
latitude            0
longitude           0
dtype: int64
```

→ The dataset contains **zero missing values** across all columns.

Checking Duplicates

```
df.duplicated().sum()
```

```
np.int64(0)
```

→ **No duplicate entries** were detected in the dataset

Column wise Data Quality Analysis

Column : Date

1. Handling Missing Values

```
df['date'].dtypes
dtype('O')
print(df['date'].isnull().sum())
0
```

2. Checking Inconsistences

```
print(df['date'].unique())
['2021-01-31' '2018-12-23' '2020-08-10' ... '2020-12-04' '2020-08-30'
 '2018-09-25']
```

Column : Country

1. Handling Missing Values

```
df['country'].dtypes
dtype('O')
print(df['country'].isnull().sum())
0
```

2. Checking Inconsistences

```
print(df['country'].unique())
['Brazil' 'India' 'Indonesia' 'United States' 'France' 'Germany' 'Greece'
 'Spain' 'Mexico' 'South Africa' 'Philippines' 'Nigeria' 'Japan' 'Canada'
 'Italy' 'Australia' 'China' 'Bangladesh' 'Chile' 'Turkey']
```

Column : Disaster type

1. Handling Missing Values

```
df['disaster_type'].dtypes  
  
dtype('O')  
  
print(df['disaster_type'].isnull().sum())  
  
0
```

2. Checking Inconsistences

```
print(df['disaster_type'].unique())  
  
['Earthquake' 'Extreme Heat' 'Hurricane' 'Wildfire' 'Flood' 'Storm Surge'  
 'Drought' 'Tornado' 'Landslide' 'Volcanic Eruption']
```

Column : severity index

1. Handling Missing Values

```
df['severity_index'].dtypes  
  
dtype('float64')  
  
print(df['severity_index'].isnull().sum())  
  
0
```

2. Checking Inconsistences

```
print(df['severity_index'].unique())  
  
[ 5.99  6.53  1.55  4.55  3.8   2.56  4.77  1.47  3.32  2.61  5.72  4.4  
  3.38  6.94  5.01  5.81  5.12  6.5   3.02  5.95  5.45  2.36  5.59  6.57  
  4.51  6.65  2.86  1.97  3.23  7.11  6.14  4.53  4.06  5.43  3.68  4.33]
```

Column : casualties

1. Handling Missing Values

```
df['casualties'].dtypes  
  
dtype('int64')  
  
print(df['casualties'].isnull().sum())  
  
0
```

2. Checking Inconsistences

```
print(df['casualties'].unique())  
  
[111 100  22  94  64  56  81  34  73 202  92  50  90  88 225 228 181  43  
 180  59  71 207 133  70  30  69  44 192  68 125 104  66 280  47  51 139  
 140  74 106 109 135 137 124  60 195 102 206  53 176 342 194 166   3  37  
  79 142  67 173  55  86 143   5  38 110 128 227  14  76 226 212 276  57]
```

Column : economic loss usd

1. Handling Missing Values

```
df['economic_loss_usd'].dtypes  
  
dtype('float64')  
  
print(df['economic_loss_usd'].isnull().sum())  
  
0
```

2. Checking Inconsistences

```
print(df['economic_loss_usd'].unique())  
  
[7934365.71 8307648.99 765136.99 ... 1805859.7 3176085.56 2933495.7 ]
```

Column : response time hours

1. Handling Missing Values

```
df['response_time_hours'].dtypes  
  
dtype('float64')  
  
print(df['response_time_hours'].isnull().sum())  
  
0
```

2. Checking Inconsistencies

```
print(df['response_time_hours'].unique())  
  
[15.62  5.03 32.54 ... 26.08 45.97 26.86]
```

Column : aid amount usd

1. Handling Missing Values

```
df['aid_amount_usd'].dtypes  
  
dtype('float64')  
  
print(df['aid_amount_usd'].isnull().sum())  
  
0
```

2. Checking Inconsistencies

```
print(df['aid_amount_usd'].unique())  
  
[271603.79 265873.81 49356.49 ... 280665.61 80331.23 354096.99]
```

Column : response efficiency score

1. Handling Missing Values

```
df['response_efficiency_score'].dtypes  
  
dtype('float64')  
  
print(df['aid_amount_usd'].isnull().sum())  
  
0
```

2. Checking Inconsistencies

```
print(df['response_efficiency_score'].unique())  
  
[83.21 96.18 60.4   ... 57.41 42.09 56.46]
```

Column : recovery days

1. Handling Missing Values

```
df['recovery_days'].dtypes  
  
dtype('int64')  
  
print(df['recovery_days'].isnull().sum())  
  
0
```

2. Checking Inconsistencies

```
print(df['recovery_days'].unique())  
  
[ 67  55  22  47  42  26  49  19  27  57  41  34  70  56  75  24  53  48  
 50  46  62  21  37  90  36  30  94  59  88  51  66  52  35  61  38  63  
 60  23  77  20  86  81  44  33  58   5  45  28  68  71  80  82  43  54  
 72  78  39  87  17  79  76 105  83  91  64  40  73  69  65  25  13  32  
 74  15 106  92  89  31  85  29   9   8  14   2 104 102  84  18  10  98  
   6  93  96 103  12  11   7  95 100   4  97  16  99   3 101 107 112 109  
108 111 110]
```

Column : latitude

1. Handling Missing Values

```
df['latitude'].dtypes  
  
dtype('float64')  
  
print(df['latitude'].isnull().sum())  
  
0
```

2. Checking Inconsistences

```
print(df['latitude'].unique())  
  
[-30.613  10.859   0.643 ... 15.217 -44.002  13.084]
```

Column : longitude

1. Handling Missing Values

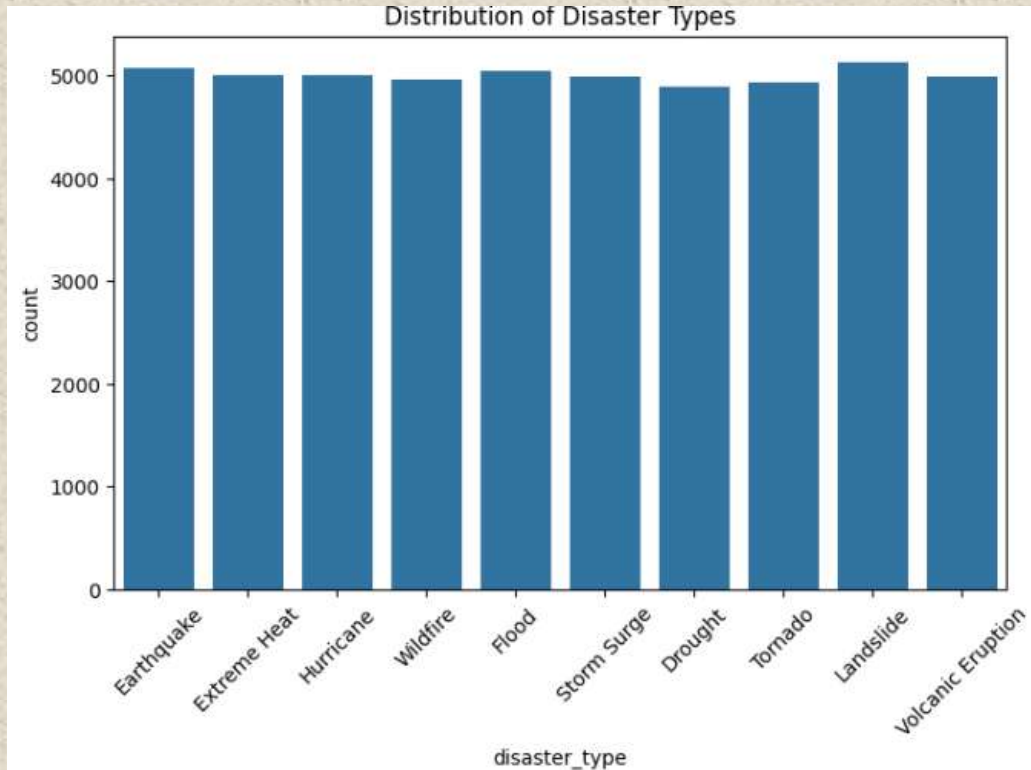
```
df['longitude'].dtypes  
  
dtype('float64')  
  
print(df['longitude'].isnull().sum())  
  
0
```

2. Checking Inconsistences

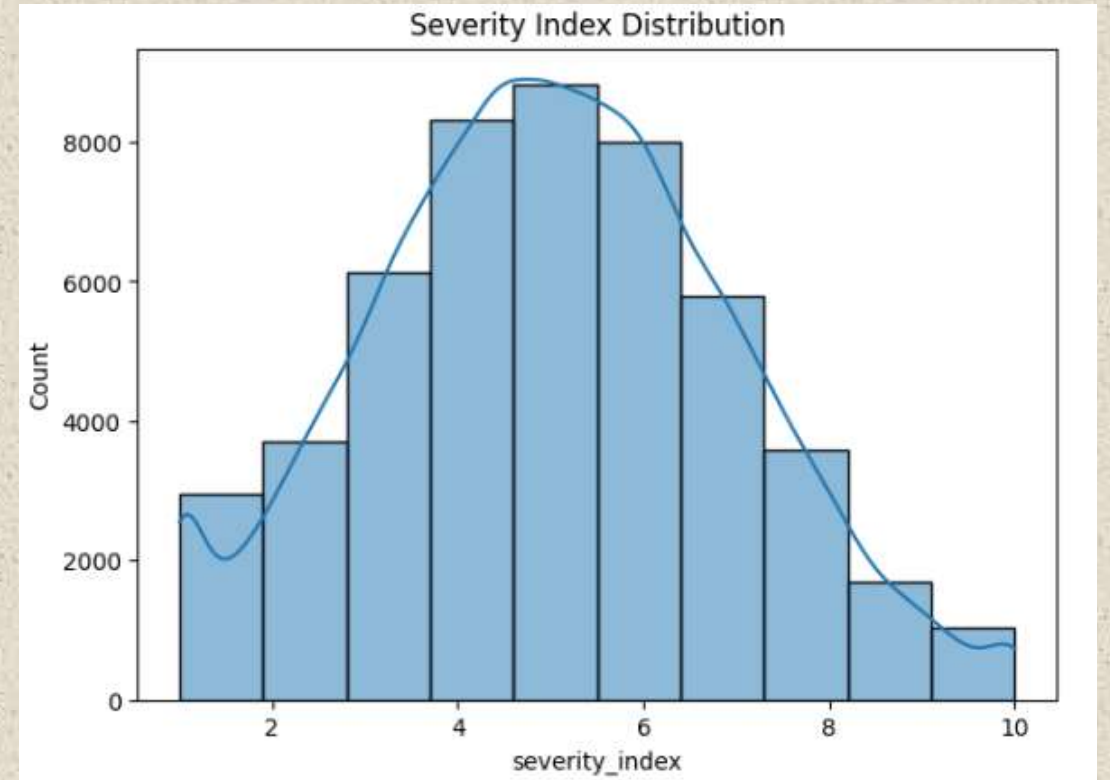
```
print(df['longitude'].unique())  
  
[-122.557 -159.194 -160.978 ... -25.68 -147.346 -27.856]
```

Data Analysis through Visualizations

Univariate Analysis

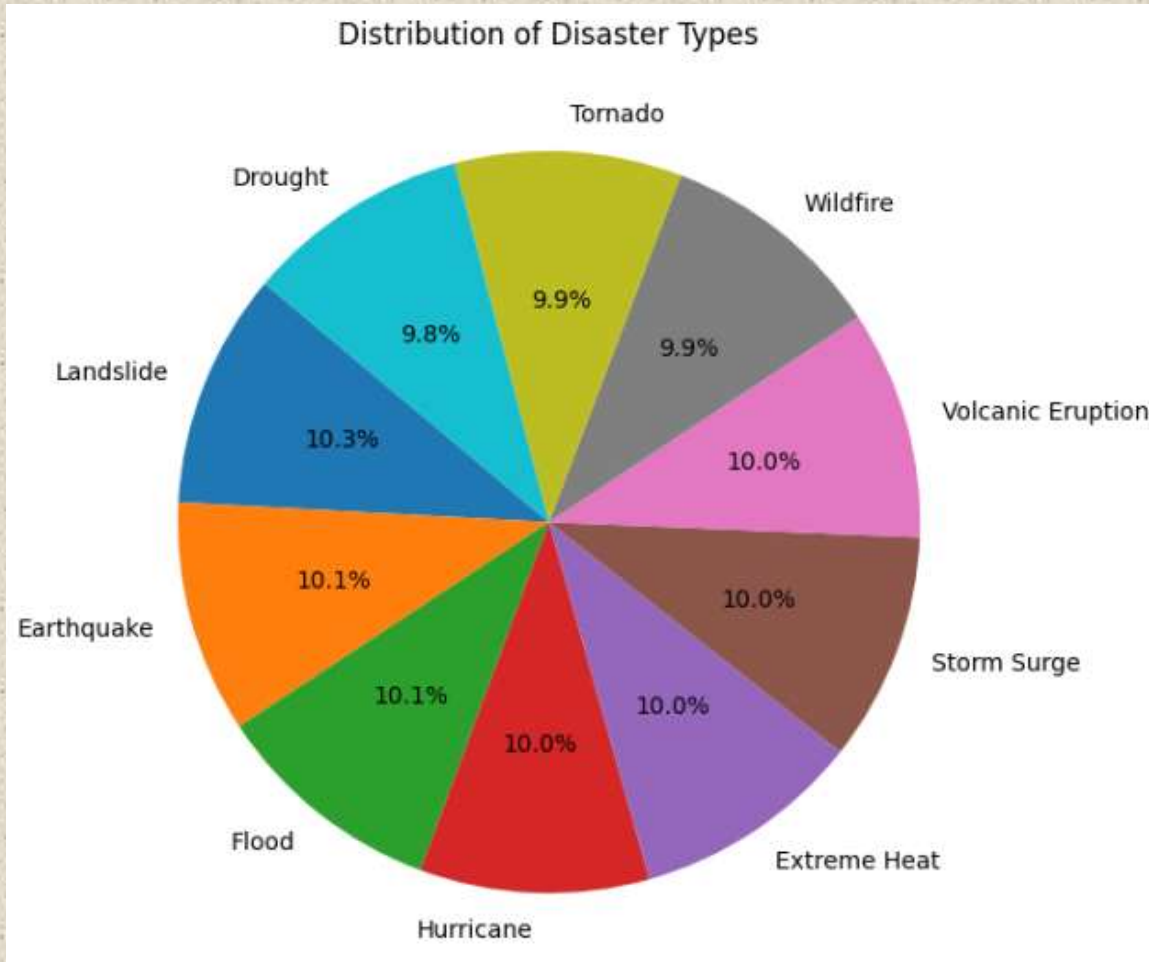


- The dataset shows an almost uniform distribution of disaster types, indicating balanced data.
- This balanced representation enables unbiased comparison across different disaster categories.



- The severity index follows a near-normal distribution, with most disasters having moderate severity levels.
- Extremely low and very high severity events are less frequent compared to mid-range values.

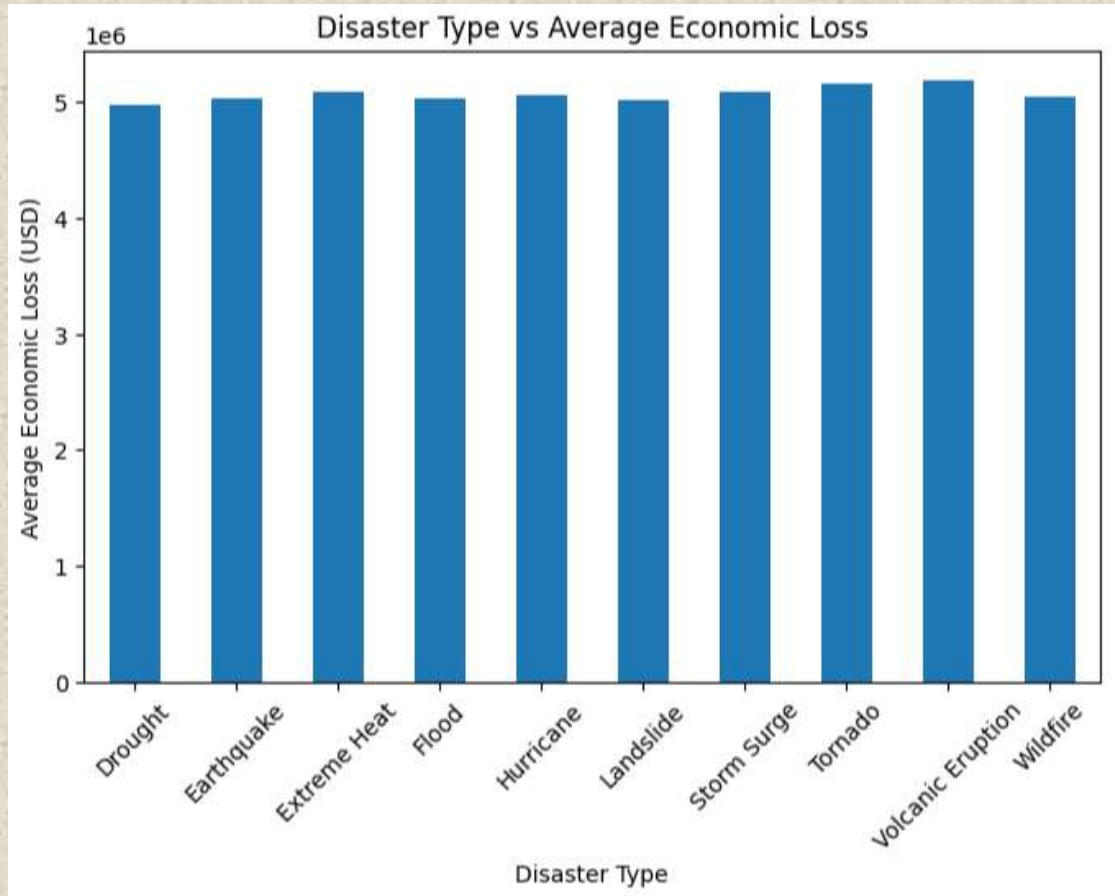
Distribution of Disaster Types



- The pie chart shows an almost equal percentage share for all disaster types, each contributing around 9–10% of total events.
- Landslides, Earthquakes, and Floods have slightly higher proportions compared to other disaster categories.
- Droughts and Tornadoes show marginally lower representation but remain comparable to other disaster types.
- The near-uniform distribution indicates a well-balanced dataset without over-representation of any single disaster type.

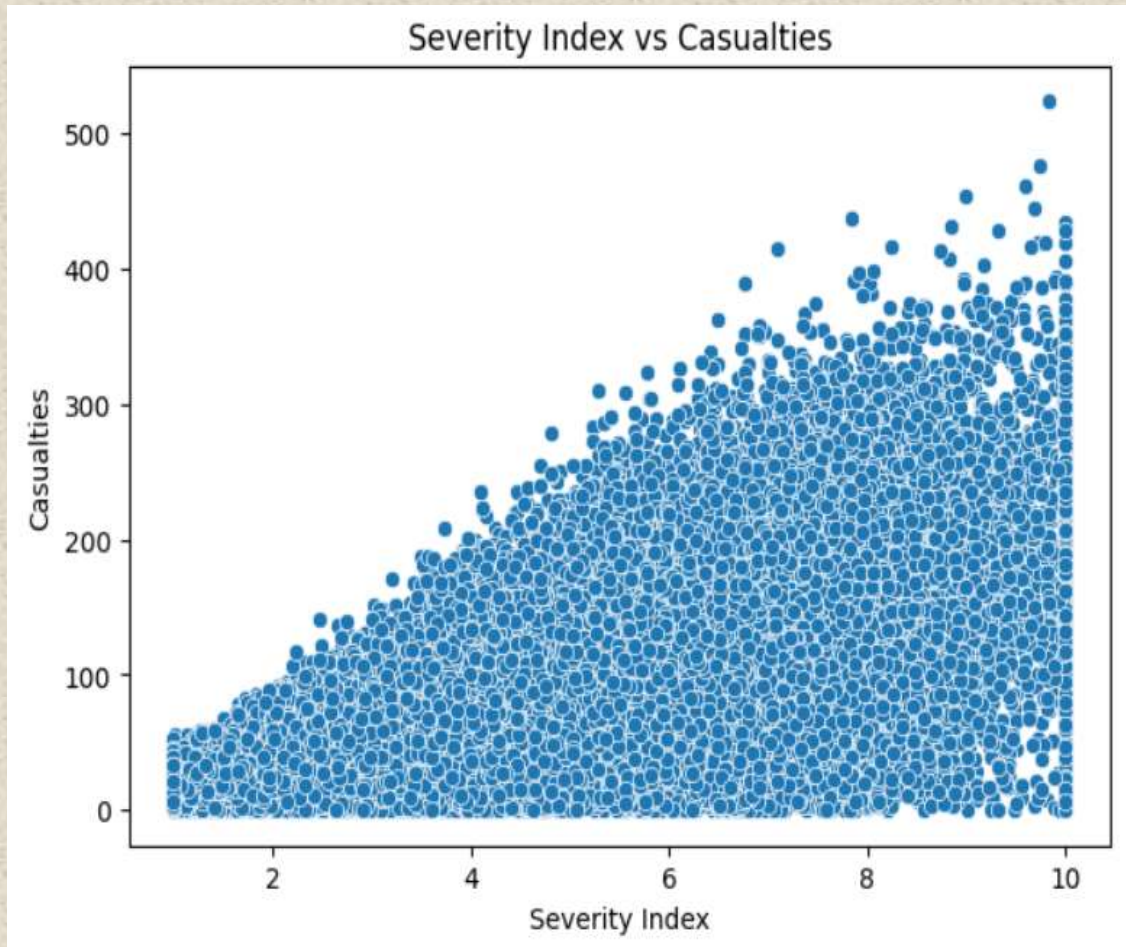
Bivariate Analysis

Categorical vs Numerical



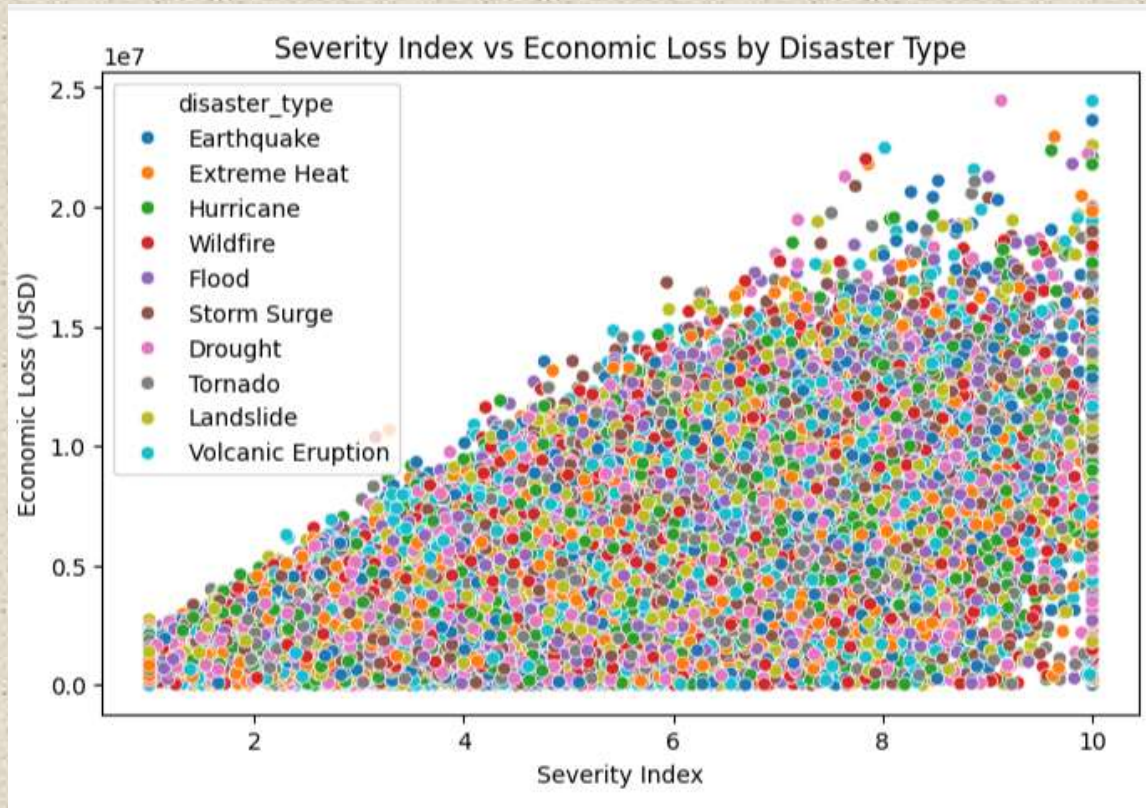
- The chart compares the average economic loss (USD) across different disaster types.
- All disaster types show high and relatively similar average losses, indicating significant financial impact across events.
- Volcanic Eruptions and Tornadoes exhibit slightly higher average economic losses compared to other disaster types.

numerical vs numerical



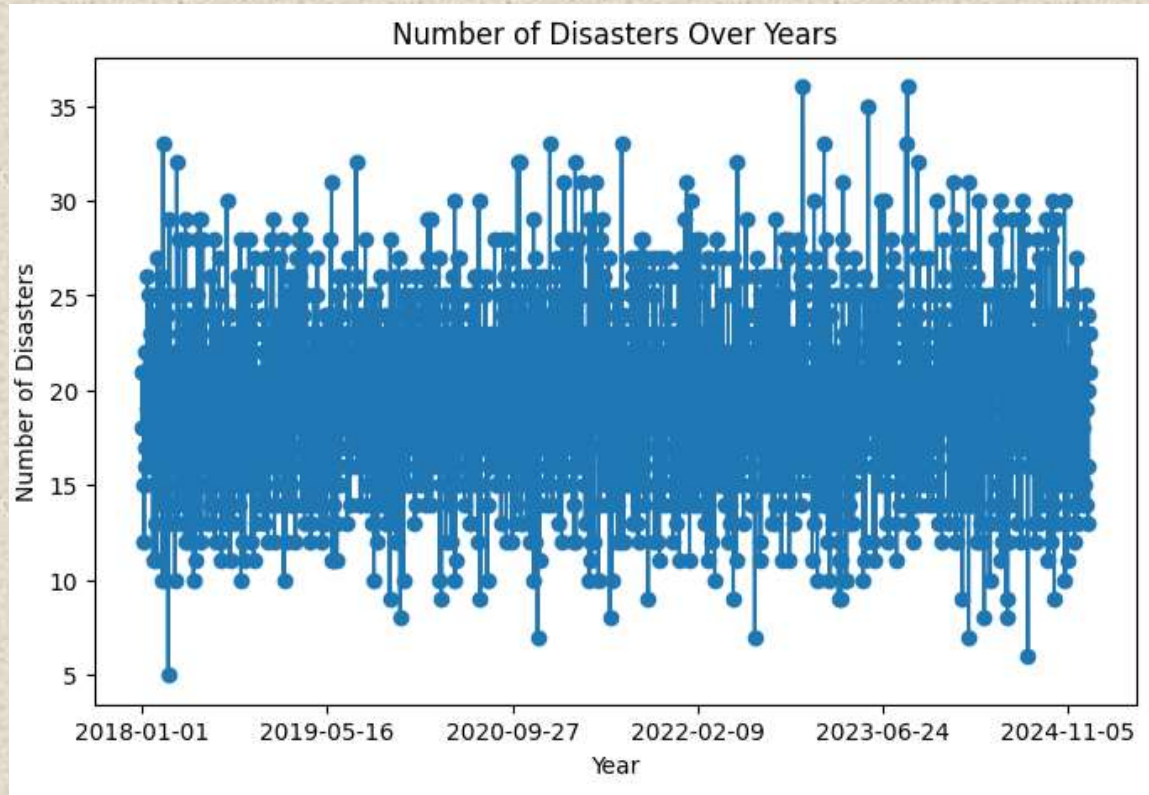
- The scatter plot shows a positive relationship between severity index and number of casualties.
- As the severity index increases, the casualties tend to increase, indicating higher impact disasters cause more losses.
- The spread of points suggests variability, meaning other factors may also influence casualties.

Multivariate Analysis



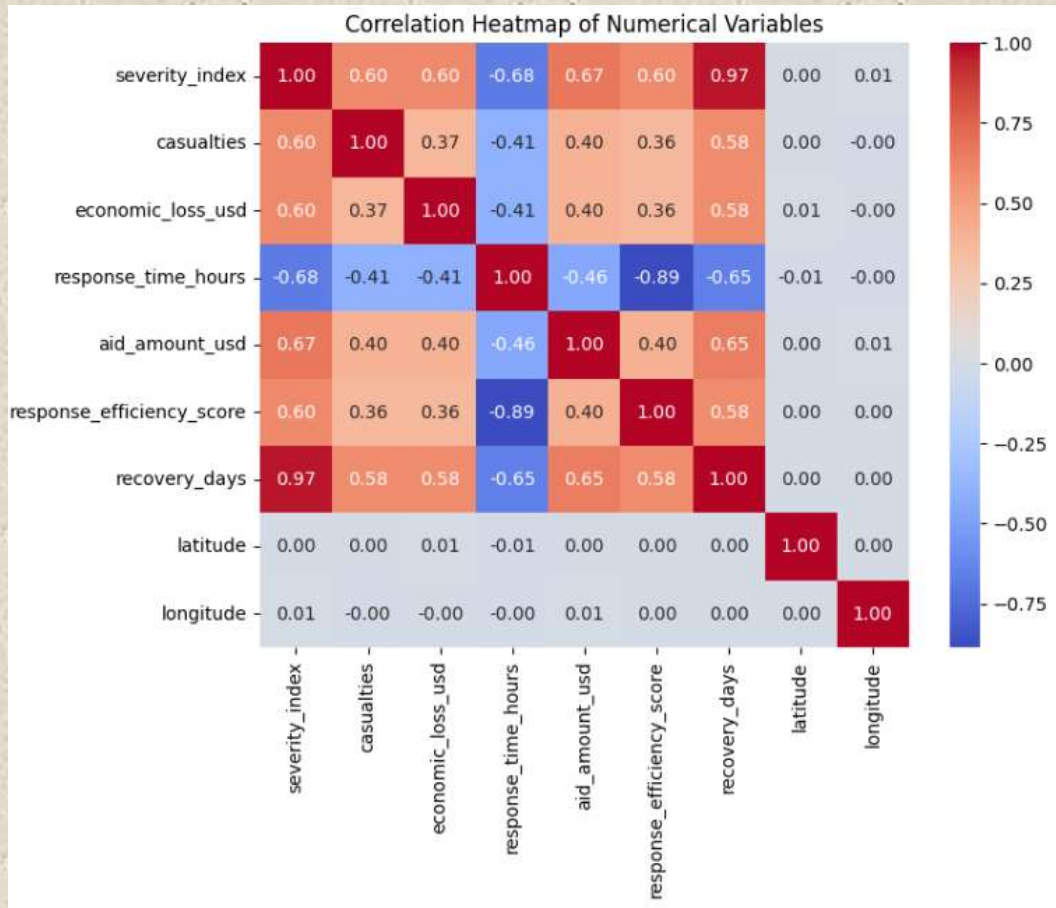
- The plot shows a positive relationship between severity index and economic loss across all disaster types.
- As disaster severity increases, the economic loss generally rises, regardless of the disaster category.
- Higher severity events (severity index above ~7) are associated with significantly larger financial impacts.
- Different disaster types follow a similar upward trend, indicating severity is a key driver of economic loss.
- The wide spread of points at higher severity levels suggests variability in impact, influenced by disaster type, location, and response effectiveness.

Time Series Analysis



- The chart shows year-wise variation in the number of disaster events from 2018 to 2024.
- Disaster occurrences fluctuate over time, indicating no single year with consistently low activity.
- A slight increasing trend can be observed in the later years, suggesting rising disaster frequency.
- Peaks in certain periods highlight years with higher disaster intensity, possibly influenced by climate and environmental factors.
- The trend emphasizes the need for continuous preparedness and long-term disaster management planning.

Correlation Heat map



- Severity index shows a strong positive correlation with recovery days, indicating that more severe disasters take longer to recover.
- Response time has a strong negative correlation with response efficiency score, meaning faster responses lead to higher efficiency.
- Casualties and economic loss are moderately positively correlated with disaster severity, showing increased impact with higher severity.
- Aid amount has a positive relationship with severity and recovery days, suggesting higher aid allocation for severe disasters.

Summary

- Analyzed global disaster response data from 2018 to 2024 covering multiple countries and disaster types.
- Performed data cleaning and preprocessing to handle inconsistencies and ensure data quality
- Conducted univariate analysis to understand disaster frequency, severity levels, casualties, and economic losses
- Used bivariate analysis to examine relationships between disaster types and key numerical impacts such as severity, recovery time, and economic loss.
- Implemented time-based analysis to identify yearly trends and seasonal patterns in disaster occurrences and impacts.
- Conducted correlation analysis to measure relationships between numerical variables and identify strong associations.
- Visualized insights using bar charts, scatter plots, heatmaps, and line graphs for better interpretation.